Global Life Expectancy & Socioeconomic Factors

A Descriptive Cross-National Data Analysis (2023 Estimates)

Topic: Public Health & Global Economics **Author:** Research & Data Analysis Group **Sample Size:** 50 Countries (5 Continents) **Date:** October 2026

1. INTRODUCTION & RESEARCH OBJECTIVES

Life expectancy at birth is one of the most fundamental indicators of national well-being, reflecting population health status, sanitary infrastructure, nutritional availability, and general social progress. While life expectancy has improved substantially over the past half-century, profound disparities remain between different geographic regions and economic classes.

The primary objective of this project is to conduct an empirical, descriptive analysis examining cross-country relationships between life expectancy and key socioeconomic drivers: national economic output (GDP per capita), public healthcare financing (% of GDP allocated to health), and human capital development (adult literacy rate). This study analyzes a standardized dataset of 50 nations to identify baseline distribution shapes, detect outlier clusters, and assess bivariate associations.

2. DATA PROVENANCE & STRUCTURE

Data were compiled and curated from official, publicly accessible multilateral research repositories:

- **World Bank Open Data** (`data.worldbank.org`): GDP Per Capita (current USD), Total Population (millions), and Adult Literacy Rate (% of population aged 15 and above).
- **World Health Organization (WHO) Global Health Observatory** (`who.int/data/gho`): Life Expectancy at birth (years) and Healthcare Expenditure (% of national GDP).

The structured dataset contains 50 complete observations across 7 standardized columns, containing zero missing values (NA count = 0), verified for immediate execution in RStudio via the companion `analysis.R` script and the delivered Excel workbook `life_expectancy_data.xlsx`.

3. COMPREHENSIVE DESCRIPTIVE STATISTICS

Parametric and non-parametric summary measures were computed across all numeric indicators.

Variable	Unit	Mean	Std. Dev	Median	Min	Max
Life Expectancy	Years	74.88	7.58	76.60	54.30	84.80
GDP Per Capita	USD ($)	24,354	26,058	12,426	492	93,457
Healthcare Expenditure	% of GDP	7.10	3.20	6.50	2.50	16.60
Adult Literacy Rate	%	86.64	16.32	95.90	47.10	99.80
Population	Millions	126.96	312.44	43.80	5.20	1,428.60

Regional Disaggregation Overview

Region	Count	Avg Life Exp	Avg GDP/Capita	Avg Health Exp (%)	Avg Literacy (%)
Oceania	2	82.7 yrs	$56,370	9.6%	99.0%
Europe	12	80.6 yrs	$47,381	9.2%	99.0%
Americas	10	77.5 yrs	$20,691	9.3%	95.6%
Asia	14	75.9 yrs	$12,298	5.1%	88.8%
Africa	12	64.4 yrs	$2,266	4.5%	68.0%

4. DISTRIBUTIONAL CHARACTERISTICS & REGIONAL DISPARITIES

An examination of numeric distribution properties reveals substantial skewness in global health outcomes. Figures 1 through 3 illustrate the global frequency distribution, dispersion patterns across geographic blocs, and rank comparisons of continental averages.

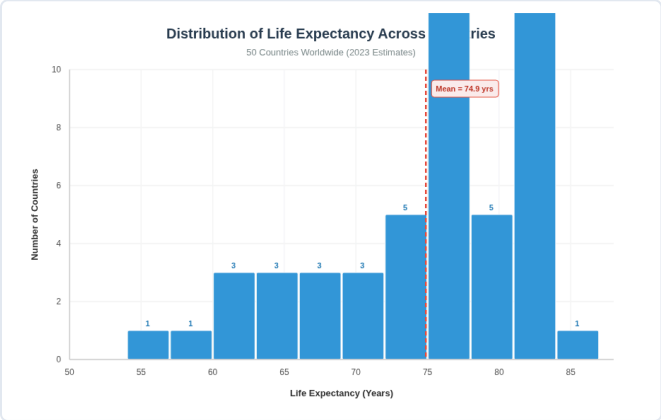


Figure 1: *Histogram of life expectancy (binwidth = 3 years). The dashed vertical line denotes the grand mean of 74.9 years. Notice the left skew with a prolonged tail.*

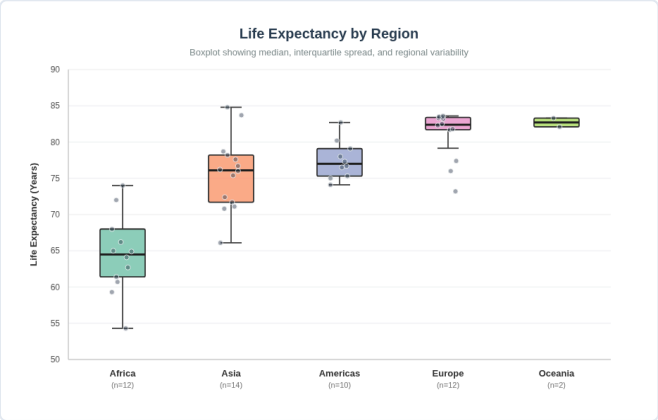


Figure 2: *Boxplot displaying median, interquartile range (IQR), and spread across regions. Jittered points display raw observations within each continental grouping.*

Distributional Properties & Outlier Diagnostics

As visualized in **Figure 1**, global life expectancy exhibits pronounced left-skewness. A high concentration of countries clusters in the 75–84 year range (mode at 81–84 years). The lower tail extends down to 54.3 years (Nigeria) and 59.3 years (Mozambique), creating a 30.5-year gap between the highest and lowest countries in the sample. The sample median (76.6 years) sits notably above the sample mean (74.88 years), a classic property of negatively skewed public health data where a minority of distressed nations pull down the arithmetic average.

The regional boxplot in **Figure 2** reveals sharp geographical stratifications:

- **Europe & Oceania:** Demonstrate high medians (>82 years) and remarkably tight interquartile spreads, indicative of universally mature healthcare systems and consistent socioeconomic safety nets.
- **Americas:** Median life expectancy reaches 77.0 years. The United States exhibits high life expectancy (79.1 years) but is paired with unprecedented healthcare spending (16.6% of GDP), behaving as an expenditure outlier.
- **Asia:** Exhibits the broadest intra-regional variance, ranging from highly developed societies (Japan at 84.8 years, South Korea at 83.7 years) down to emerging economies (Pakistan at 66.1 years).
- **Africa:** Records both the lowest regional median (63.4 years) and the highest internal dispersion (standard deviation = 5.56 years), with northern states (Morocco at 74.0 yrs) performing substantially above sub-Saharan nations.

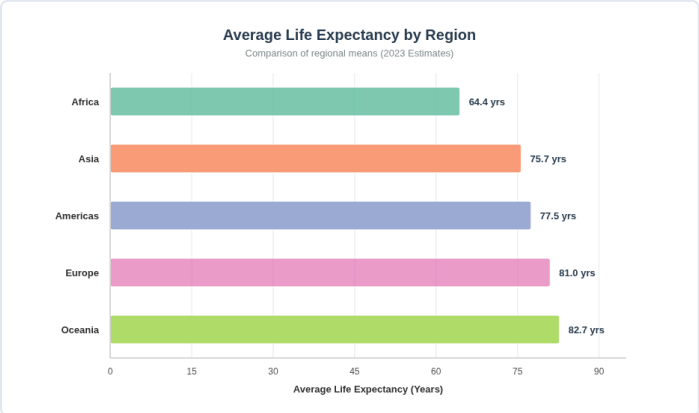


Figure 3: *Horizontal bar chart of regional mean life expectancies. A distinct 18.3-year gap separates Oceania from Africa.*

Regional Gap Summary

Figure 3 summarizes the stark continental divide: Europe (80.6 yrs) and Oceanian (82.7 yrs) citizens live on average to 18 years longer than citizens in African nations (64.4 yrs).

> **Key Observation:** The regional difference is strongly paired with economic resources: the average European GDP/capita ($47,381) in the sample is more than 20 times greater than that of the sampled African nations ($2,266).

5. SOCIOECONOMIC ASSOCIATIONS & CORRELATION ANALYSIS

Bivariate relationships were evaluated between life expectancy and economic prosperity, healthcare expenditure, and adult literacy. All three factors show positive, statistically robust associations.

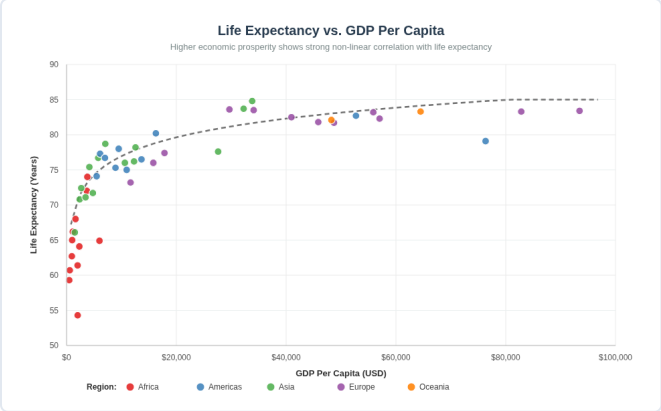


Figure 4: Scatter plot of Life Expectancy vs. GDP Per Capita with logarithmic trend curve (r = 0.694). Diminishing returns appear above $30,000.

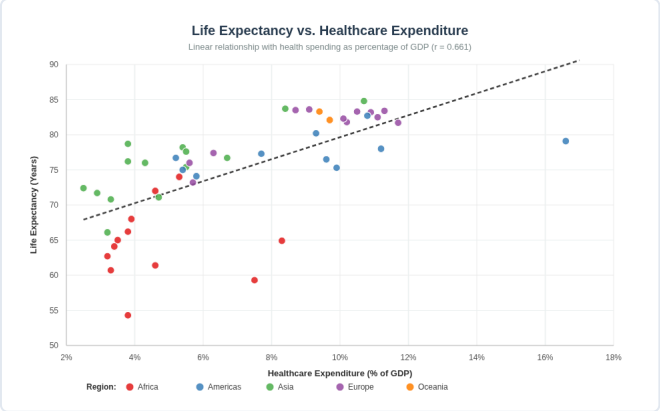


Figure 5: Life Expectancy vs. Healthcare Expenditure as % of GDP with linear fit (r = 0.661). Higher spending aligns with longer life.

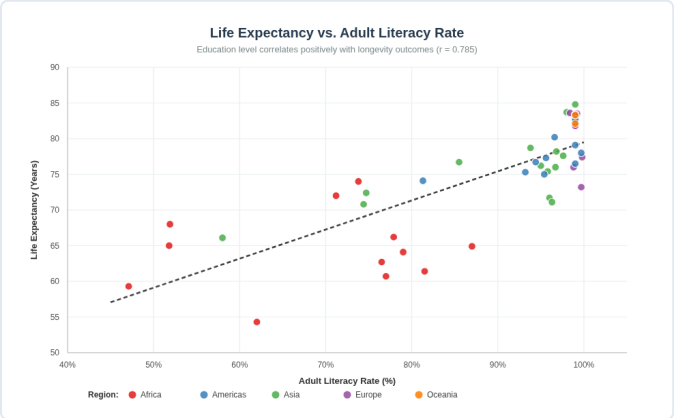


Figure 6: Life Expectancy vs. Adult Literacy Rate with linear fit (r = 0.785). Strongest single linear bivariate predictor in the dataset.

Comparative Correlation Matrix

Indicator Pair	Pearson r	Relationship Strength
Life Exp vs. Adult Literacy	**+0.785**	Strong positive
Life Exp vs. GDP Per Capita	**+0.694**	Moderate-to-strong non-linear
Life Exp vs. Healthcare Exp %	**+0.661**	Moderate positive

The Preston Curve Effect: In **Figure 4**, the relationship with GD per capita is distinctly concave. At early development stages (GD < $10,000), marginal economic increases correspond to ste gains in lifespan. Beyond $35,000, longevity plateaus between and 85 years.

6. DISCUSSION & METHODOLOGICAL SCOPE

The findings corroborate established developmental economics: basic income, sanitary infrastructure, and general literacy constitute essential foundations for maternal health and infectious disease containment. Literacy (r = 0.785) shows the highest direct linear correlation with longevity, reflecting how maternal education and health literacy directly influence community survival.

> **Methodological Scope & Causal Disclaimer:** This study is strictly descriptive and cross-sectional. The observed positive correlations (r = 0.66 to 0.79) do not establish causal relationships. Confounding factors—such as institutional quality, environmental conditions, and peacetime stability—simultaneously influence both economic metrics and health outcomes.

7. CONCLUSIONS

- **Socioeconomic Underpinnings:** Life expectancy is tightly bound to national economic capacity and human development, with literacy rate being the strongest linear covariate.
- **Non-Linear Economic Returns:** Wealth yields massive life gains in developing nations, but exhibits diminishing marginal returns in high-income states.
- **Closing the Global Gap:** Addressing the 18.3-year lifespan disparity between Africa and advanced economies requires integrated investments in both public healthcare capacity and foundational education.